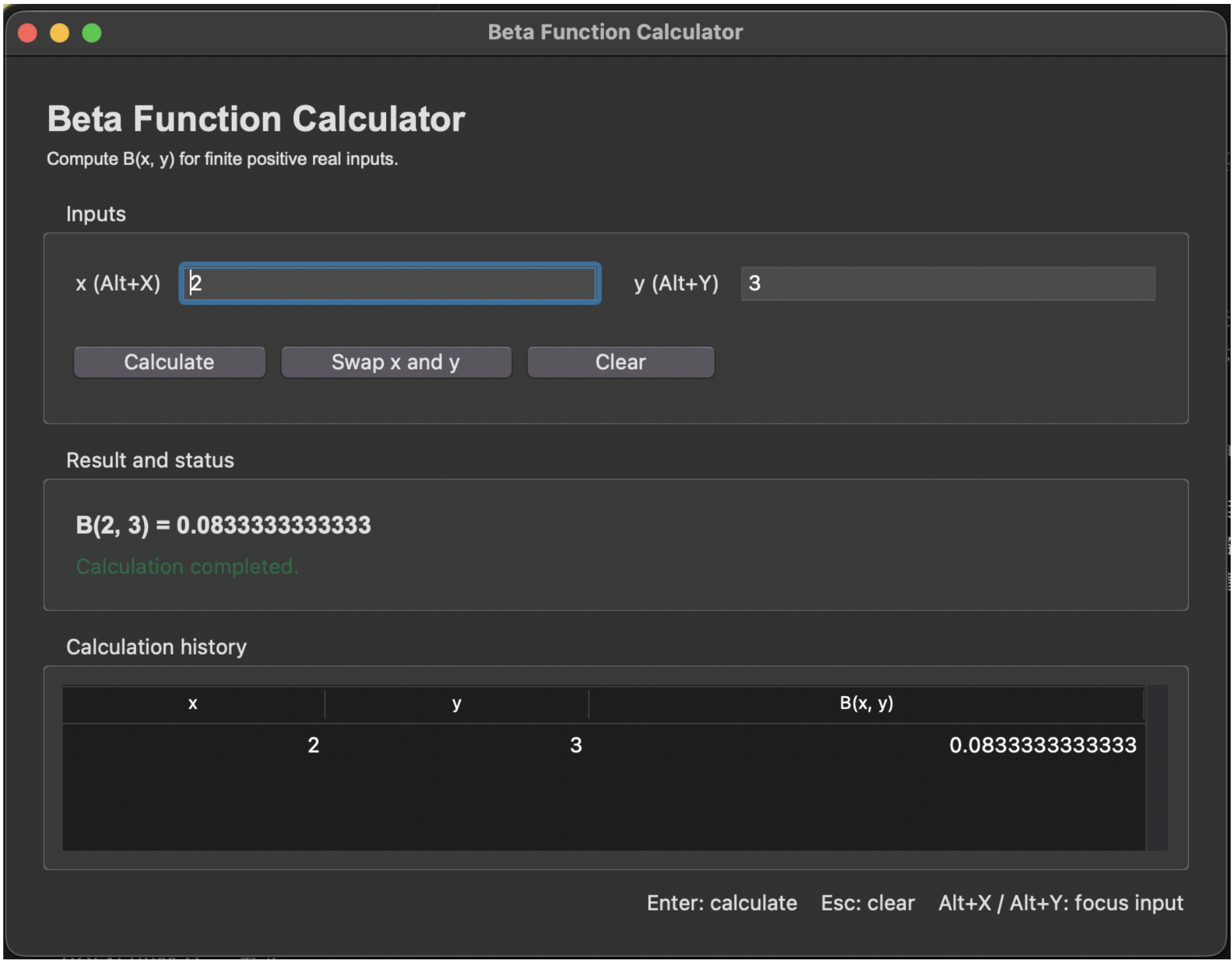


$$B(x,y) = \frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)}, \quad x > 0, y > 0$$

Final Graphical Interface



The live interface demonstrates a known value, visible calculation status, and persistent history. Native Tkinter widgets preserve familiar focus and tab behavior.

Accessibility and Recovery

- Keyboard access

Enter, Esc, Alt+X, Alt+Y
- Recognition over recall

visible labels and actions
- Helpful recovery

field-specific text messages
- Status is not color-only

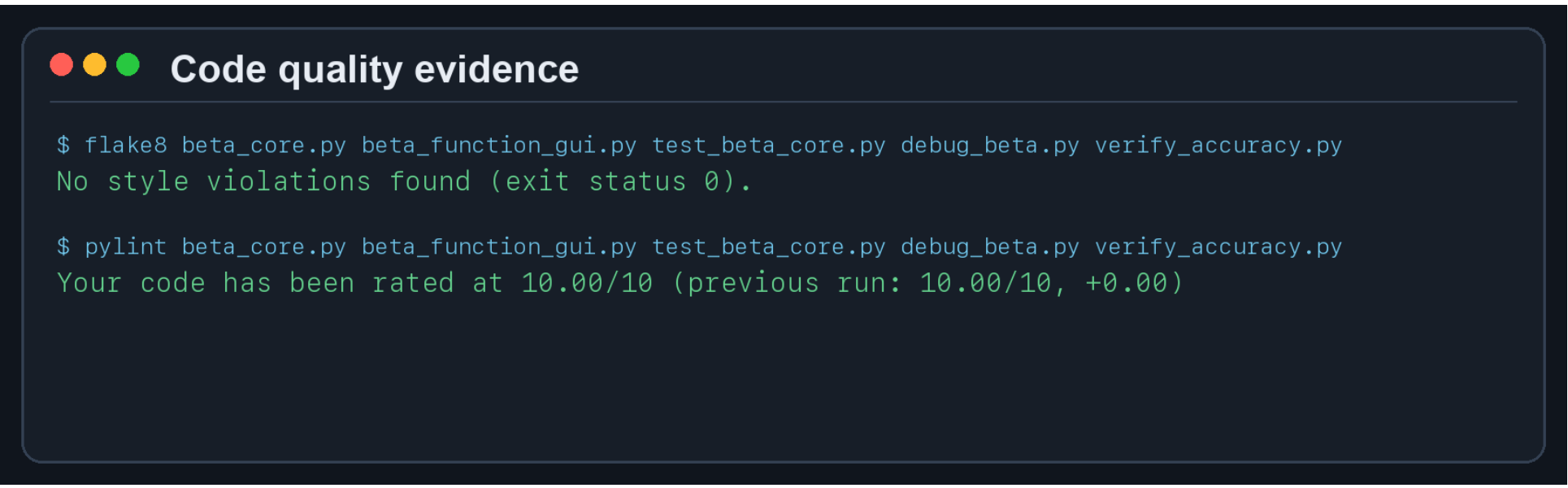
words identify success or error

The GUI aims to be accessible through readable labels, visible focus, redundant text feedback, key-board operation, and recoverable validation. These choices align the interface with the user’s task model (Kamthan, 2026c).

Exception policy: expected input and numeric-range errors receive specific messages; unexpected arithmetic failures are not hidden behind a generic message and remain available for diagnosis.

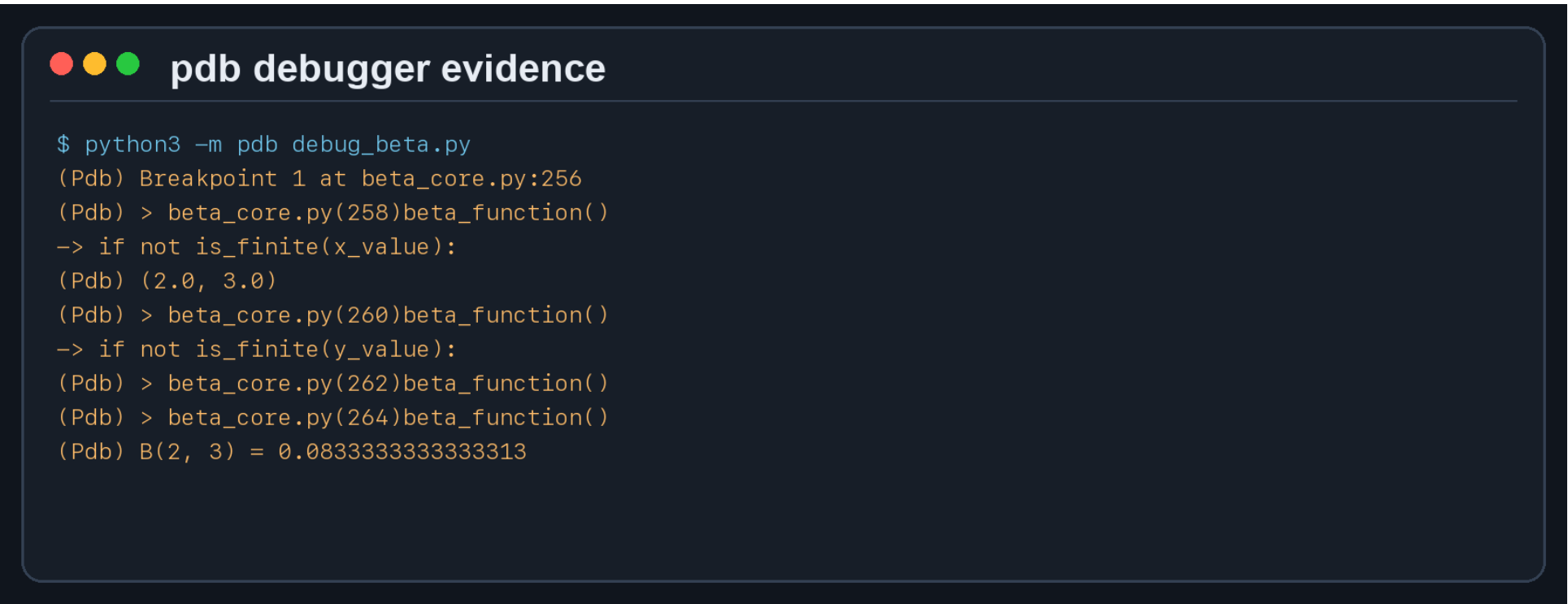
D3 conclusion: the release improves numerical stability, accessibility, and software quality through a stable log-beta implementation, explicit range behavior, and repeatable verification.

Problem 7: Style and Static Analysis



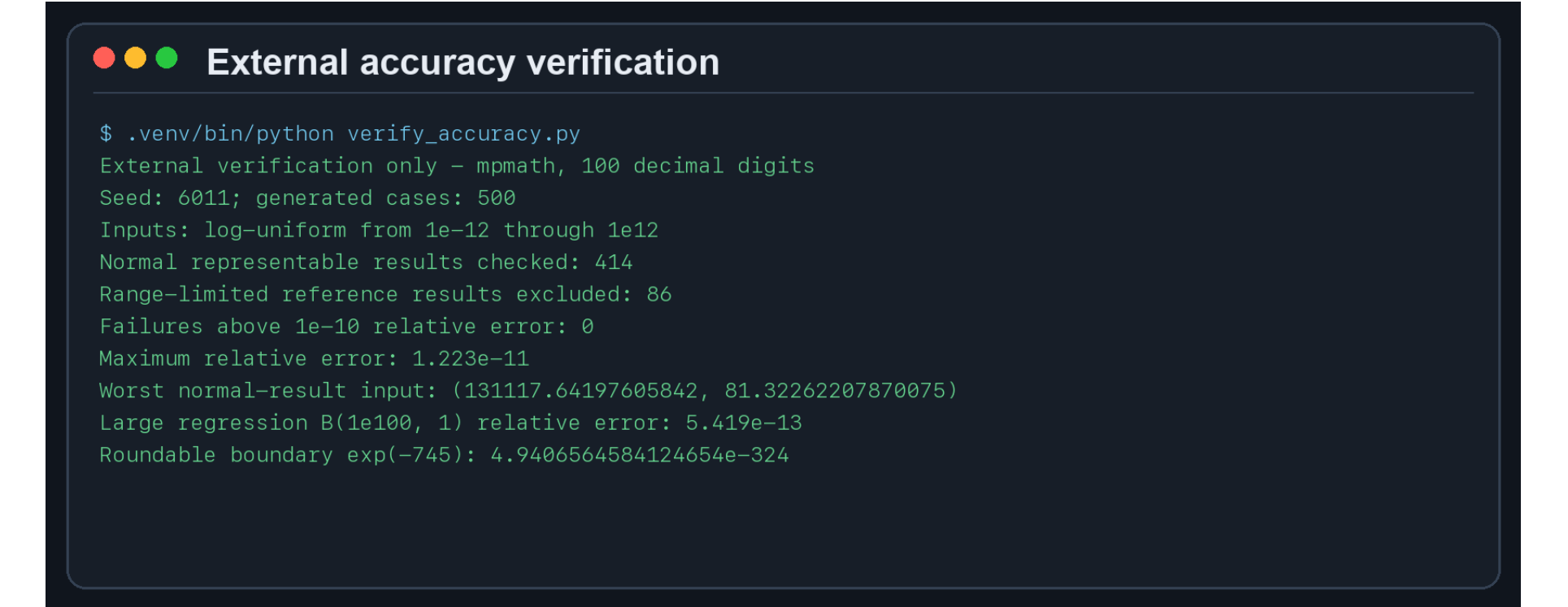
Flake8 reports no PEP 8 style violations; Pylint rates the final source **10.00/10**. Both snapshots were generated from version 1.0.0.

Problem 7: Debugger Evidence



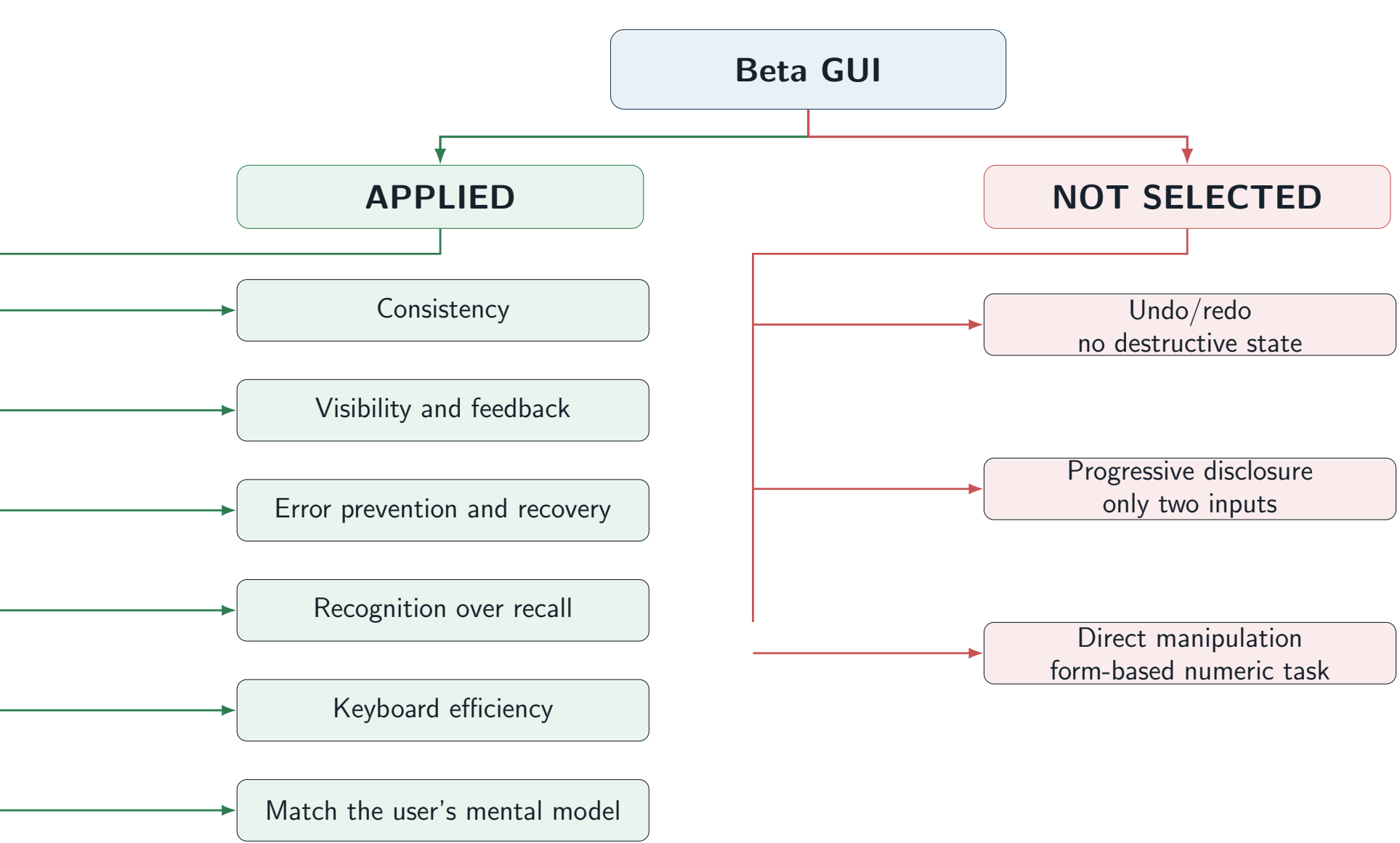
A reproducible pdb session stops inside beta_function, inspects (x,y) = (2,3), steps through validation, and confirms the known result 1/12.

Verified Numerical Contract



Contract: finite positive inputs produce a representable binary64 result or a specific range error. Twelve displayed digits are formatting, not a universal accuracy claim; subnormal values follow binary64 spacing.

UIDP Decision Mind Map



The mind map records both selected and rejected principles instead of treating every principle as mandatory (Kamthan, 2026d).

Problem 8: PyUnit Test Suite



14/14 tests pass. Coverage includes trusted values, symmetry, extreme asymmetric inputs, the round-able subnormal boundary, scientific notation, invalid input, underflow, and propagation of unexpected arithmetic errors. The tests exercise the final numerical core and parser.

Generative AI and CASTROFF. Tool: OpenAI ChatGPT/Codex. Problem 7 prompt: “Review the stable log-beta code for PEP 8, Flake8, pdb, Pylint, SemVer, UIDP, accessibility, and honest range behavior.” Problem 8 prompt: “Propose PyUnit regressions for large inputs, subnormal rounding, invalid input, and unexpected exceptions.” Context: SOEN 6011 D3; Audience: TA/instructor; Style: concise; Task: review; Role: student engineer; Output: suggestions; Format: poster checklist; Feedback: accept only after execution. Suggestions were revised and verified; unverified claims were rejected.

References. Kamthan, P. (2026a), *SOEN 6011 Project Description*. Kamthan, P. (2026b), *Building, Versioning, and Releasing Software*. Kamthan, P. (2026c), *Introduction to Mental Models*. Kamthan, P. (2026d), *Brainstorming and Mind Mapping*. Van Rossum, G., Warsaw, B., and Coghlan, N. (2001), *PEP 8 – Style Guide for Python Code*. Python Software Foundation (2026), *unittest and pdb Documentation*.

Repository: github.com/HW925/SOEN-6011-D3-Beta-Function